

Homework 3 Solution

Problem 1. Define key-generation algorithm $\mathcal{K}$ to output a random 128-bit key K and define encryption algorithm $\mathcal{E}$ by

Algorithm $\mathcal{E}_K(M)$:
 $C[0] \leftarrow_{\$} \{0, 1\}^{128}$
 For $i = 1$ to m do:
 $W[i] \leftarrow C[0] + i \bmod 2^{128}$
 $C[i] \leftarrow \text{AES}_K(M[i] \oplus W[i])$
 $C \leftarrow C[0] \parallel \dots \parallel C[m]$
 Return C

Above we parse M as consisting of m blocks of 128-bits each, and ' $W[i] \leftarrow C[0] + i \bmod 2^{128}$ ' denotes regarding $C[0]$ and i as encoding 128-bit integers, taking their sum modulo 2^{128} , and then encoding the result as another 128-bit string $W[i]$.

(Part A.) Define a decryption algorithm $\mathcal{D}$ such that $\text{SE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ is a symmetric-key encryption scheme (i.e., satisfying the correctness condition we gave in class).

Solutions:

The decryption algorithm is defined as the following:

Algorithm $\mathcal{D}_K(C)$:
 $C[0] \parallel \dots \parallel C[m] \leftarrow C$, where $|C[0]| = \dots = |C[m]| = 128$
 For $i = 1$ to m do:
 $W[i] \leftarrow C[0] + i \bmod 2^{128}$
 $M[i] \leftarrow \text{AES}_K^{-1}(C[i]) \oplus W[i]$
 $M \leftarrow M[1] \parallel \dots \parallel M[m]$
 Return M

(Part B.) Show that SE is not IND-CPA secure.

Solution:

The adversary is defined as:

Adversary A :

$M_L \leftarrow 0^{127}1||0^{128}$

$M_R \leftarrow 0^{256}$

$C \leftarrow \text{LR}(M_L, M_R)$

$C[0]||C[1]||C[2] \leftarrow C$

If $C[1] = C[2]$ then return 0

Else return 1

For the running-time analysis, note that A makes only 1 LR-query and does minor additional computation on 128-bit blocks, so is extremely practical. Now, we claim that

$$\text{Adv}_{\text{SE}}^{\text{ind-cpa}}(A) \geq 1/4 .$$

Now by definition

$$\text{Adv}_{\text{SE}}^{\text{ind-cpa}}(A) = \Pr[\text{Right}_{\text{SE}}^A = 1] - \Pr[\text{Left}_{\text{SE}}^A = 1]$$

so we conclude via the following claims.

Claim: $\Pr[\text{Right}_{\text{SE}}^A = 1] = 1$.

Proof. In the right game, $C \leftarrow \mathcal{E}_K(M_R)$. $M_R = 0^{256}$, we know that:

$$C[1] = \text{AES}_K(C[0] + 1 \bmod 2^{128}) \text{ and } C[2] = \text{AES}_K(C[0] + 2 \bmod 2^{128})$$

Recall that AES is a blockcipher and blockciphers are permutations in that under the same key each plaintext maps to a *different* output block. Thus $C[0] + 1 \bmod 2^{128} \neq C[0] + 2 \bmod 2^{128}$ implies that $\text{AES}_K(C[0] + 1 \bmod 2^{128}) \neq \text{AES}_K(C[0] + 2 \bmod 2^{128})$, that is $C[1] \neq C[2]$. It follows that in the right game, A always outputs 1, meaning that $\Pr[\text{Right}_{\text{SE}}^A = 1] = 1$. $\square$

Claim: $\Pr[\text{Left}_{\text{SE}}^A = 1] \leq \frac{3}{4}$.

Proof. In the left game $C \leftarrow \mathcal{E}_K(M_L)$. Note that there is 1/4 chance that the last two bits of $C[0]$ is 01 as $C[0]$ are random. For the remainder of the proof we condition on this being the case, and let $X||00 \leftarrow C[0]$, where $|X| = 126$. Then $W[1] = X||10$ and $W[2] = X||11$. So

$$W[1] \oplus 0^{127}||1 = W[2] \oplus 0^{128} = X||11 .$$

Again using that AES is a blockcipher, $\text{AES}_K(W[1] \oplus 0^{127}||1) = \text{AES}_K(W[2] \oplus 0^{128})$, that is $C[1] = C[2]$. This implies that A has at least 1/4 chance of outputting 0 in the left game. $\square$